

DAY 19 CODING CHALLENGE

```
CREATE DATABASE agri_innovate;
```

```
USE agri_innovate;
```

```
CREATE TABLE farmers (  
    farmer_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    email VARCHAR(100),  
    hire_date DATE  
);
```

```
CREATE TABLE plots (  
    plot_id INT PRIMARY KEY,  
    plot_name VARCHAR(100),  
    farmer_id INT,  
    crop_type VARCHAR(50),  
    soil_type VARCHAR(50),  
    FOREIGN KEY (farmer_id) REFERENCES farmers(farmer_id)  
);
```

```
CREATE TABLE yields (  
    yield_id INT PRIMARY KEY,  
    plot_id INT,  
    harvest_date DATE,  
    yield_kg DECIMAL(10,2),  
    weather_condition VARCHAR(50),
```

```
FOREIGN KEY (plot_id) REFERENCES plots(plot_id)
);
```

```
CREATE TABLE irrigation_logs (
    log_id INT PRIMARY KEY,
    plot_id INT,
    irrigation_date DATE,
    water_amount_liters DECIMAL(10,2),
    FOREIGN KEY (plot_id) REFERENCES plots(plot_id)
);
```

```
INSERT INTO farmers VALUES
```

```
(1, 'Ravi', 'Kumar', 'ravi@example.com', '2022-01-10'),
(2, 'Sita', 'Reddy', 'sita@example.com', '2021-03-15'),
(3, 'Arjun', 'Naidu', 'arjun@example.com', '2020-07-20');
```

```
INSERT INTO plots VALUES
```

```
(101, 'West Field', 1, 'Wheat', 'Loam'),
(102, 'North Pasture', 2, 'Corn', 'Clay'),
(103, 'South Ridge', 3, 'Soybeans', 'Sand'),
(104, 'East Field', 1, 'Wheat', 'Clay');
```

```
INSERT INTO yields VALUES
```

```
(1, 101, '2025-01-10', 5000, 'Sunny'),
(2, 101, '2025-03-10', 5200, 'Mild'),
(3, 102, '2025-02-15', 4800, 'Rainy'),
(4, 103, '2025-03-20', 4500, 'Sunny'),
```

(5, 104, '2025-02-25', 4000, 'Rainy'),

(6, 104, '2025-03-30', 4200, 'Mild');

INSERT INTO irrigation_logs VALUES

(1, 101, '2025-01-01', 1500),

(2, 101, '2025-02-01', 1600),

(3, 102, '2025-01-10', 1400),

(4, 103, '2025-02-05', 1300),

(5, 104, '2025-01-15', 1800),

(6, 104, '2025-02-20', 1700);

66

67 • `SELECT * FROM farmers;`

68

Result Grid   Filter Rows: Edit:    Export/Import

	farmer_id	first_name	last_name	email	hire_date
▶	1	Ravi	Kumar	ravi@example.com	2022-01-10
	2	Sita	Reddy	sita@example.com	2021-03-15
	3	Arjun	Naidu	arjun@example.com	2020-07-20
✱	NULL	NULL	NULL	NULL	NULL

69 • `SELECT * FROM plots;`

70

71 • `SELECT * FROM yields;`

Result Grid					
				Filter Rows:	Edit:   
	plot_id	plot_name	farmer_id	crop_type	soil_type
▶	101	West Field	1	Wheat	Loam
	102	North Pasture	2	Corn	Clay
	103	South Ridge	3	Soybeans	Sand
	104	East Field	1	Wheat	Clay
•	NULL	NULL	NULL	NULL	NULL

71 • `SELECT * FROM yields;`

72

73 • `SELECT * FROM irrigation_logs;`

Result Grid					
				Filter Rows:	Edit:   
	yield_id	plot_id	harvest_date	yield_kg	weather_condition
▶	1	101	2025-01-10	5000.00	Sunny
	2	101	2025-03-10	5200.00	Mild
	3	102	2025-02-15	4800.00	Rainy
	4	103	2025-03-20	4500.00	Sunny
	5	104	2025-02-25	4000.00	Rainy
	6	104	2025-03-30	4200.00	Mild
•	NULL	NULL	NULL	NULL	NULL

```
73 • SELECT * FROM irrigation_logs;
```

```
74
```

Result Grid				
Filter Rows:				
Edit: 				
	log_id	plot_id	irrigation_date	water_amount_liters
▶	1	101	2025-01-01	1500.00
	2	101	2025-02-01	1600.00
	3	102	2025-01-10	1400.00
	4	103	2025-02-05	1300.00
	5	104	2025-01-15	1800.00
	6	104	2025-02-20	1700.00
✱	NULL	NULL	NULL	NULL

-- Task 1.1: Top 3 Most Productive Plots

```
SELECT
  p.plot_name,
  p.crop_type,
  AVG(y.yield_kg) AS average_yield_kg
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
GROUP BY p.plot_id, p.plot_name, p.crop_type
ORDER BY average_yield_kg DESC
LIMIT 3;
```

80 n crop type

Result Grid |   Filter Rows: | 

	plot_name	crop_type	average_yield_kg
▶	West Field	Wheat	5100.000000
	North Pasture	Corn	4800.000000
	South Ridge	Soybeans	4500.000000

-- Task 1.2: Total Water Consumption per Plot

```

SELECT
    p.plot_name,
    SUM(i.water_amount_liters) AS total_water_liters
FROM plots p
JOIN irrigation_logs i ON p.plot_id = i.plot_id
GROUP BY p.plot_id, p.plot_name
ORDER BY total_water_liters DESC;

```

Result Grid |   Filter Rows:

	plot_name	total_water_liters
▶	East Field	3500.00
	West Field	3100.00
	North Pasture	1400.00
	South Ridge	1300.00

-- Task 2.1: Average Yield per Crop & Weather

```

SELECT
    p.crop_type,
    y.weather_condition,
    AVG(y.yield_kg) AS average_yield_kg
FROM yields y
JOIN plots p ON y.plot_id = p.plot_id
GROUP BY p.crop_type, y.weather_condition
ORDER BY p.crop_type, y.weather_condition;

```

Result Grid			
Filter Rows:			
	crop_type	weather_condition	average_yield_kg
▶	Corn	Rainy	4800.000000
	Soybeans	Sunny	4500.000000
	Wheat	Mild	4700.000000
	Wheat	Rainy	4000.000000
	Wheat	Sunny	5000.000000

- Task 2.2: Highest Yield per Soil Type

```

SELECT soil_type, plot_name, highest_yield_kg
FROM (
    SELECT
        p.soil_type,
        p.plot_name,
        MAX(y.yield_kg) AS highest_yield_kg,
        ROW_NUMBER() OVER (

```

```

PARTITION BY p.soil_type
ORDER BY MAX(y.yield_kg) DESC
) AS rn
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
GROUP BY p.soil_type, p.plot_name
) ranked
WHERE rn = 1;

```

Result Grid			
Filter Rows:			
	soil_type	plot_name	highest_yield_kg
▶	Clay	North Pasture	4800.00
	Loam	West Field	5200.00
	Sand	South Ridge	4500.00

-- Task 3.1: Farmer with Lowest Average Water Usage

```

SELECT
    f.first_name,
    f.last_name,
    AVG(i.water_amount_liters) AS average_water_liters_per_plot
FROM farmers f
JOIN plots p ON f.farmer_id = p.farmer_id
JOIN irrigation_logs i ON p.plot_id = i.plot_id
GROUP BY f.farmer_id, f.first_name, f.last_name
ORDER BY average_water_liters_per_plot ASC
LIMIT 1;

```


Result Grid				Filter Rows:	Export:
	first_name	last_name	average_water_liters_per_plot		
	Arjun	Naidu	1300.000000		

- Task 3.2: Harvest Count per Month (Last 12 Months)

Result Grid			Filter Rows:
	month	number_of_harvests	
▶	2025-01	1	
	2025-02	2	
	2025-03	3	

-- Task 4: Advanced Analysis (Bonus)

```

WITH avg_yield AS (
    SELECT
        p.crop_type,
        AVG(y.yield_kg) AS avg_yield
    FROM plots p
    JOIN yields y ON p.plot_id = y.plot_id
    GROUP BY p.crop_type
),
avg_water AS (
    SELECT
        p.crop_type,
        AVG(i.water_amount_liters) AS avg_water

```

```

FROM plots p
JOIN irrigation_logs i ON p.plot_id = i.plot_id
GROUP BY p.crop_type
)

```

```

SELECT
    p.plot_name,
    p.crop_type,
    y.yield_kg,
    i.water_amount_liters
FROM plots p
JOIN yields y ON p.plot_id = y.plot_id
JOIN irrigation_logs i ON p.plot_id = i.plot_id
JOIN avg_yield ay ON p.crop_type = ay.crop_type
JOIN avg_water aw ON p.crop_type = aw.crop_type
WHERE
    y.yield_kg < ay.avg_yield
    AND i.water_amount_liters > aw.avg_water;

```

	plot_name	crop_type	yield_kg	water_amount_liters
►	East Field	Wheat	4000.00	1800.00
	East Field	Wheat	4000.00	1700.00
	East Field	Wheat	4200.00	1800.00
	East Field	Wheat	4200.00	1700.00

THANK YOU

HEMALATHA

